

Education

Doctor of Philosophy	Supervisor: Prof. Ankush Sharma	2021-2026	
Indian Institute of Technology, Kanpur			Uttar Pradesh India
Power Engineering			
9.2 CGPA			
Masters	Supervisor: Prof. Ankush Sharma	2020-2021	
Indian Institute of Technology, Kanpur			Uttar Pradesh India
Power Engineering			
8.5 CGPA			
Bachelor of Technology		2014-2018	
College of Technology and Engineering, Udaipur			Rajasthan India
Electrical Engineering			

Research Internship

SGREAL- lab	Supervisor: Prof. Anurag Srivastava	2023	
West Verginia University (WVU), Morganton			West Verginia USA
Area: Edge-Driven Distributed Control for Power Distribution			

Research Area

Development of micro-Phasor Measurement Unit (u-PMU).

Signal Processing, Sparse Decomposition and Synchro phasor Estimation.

Cyber and Physical security challenges in wide area monitoring system for power distribution system.

High Speed 5G communication for real time monitoring using PMUs and uPMUs.

Integration of Indian Satellite constellation system (NavIC) for time synchronization in uPMU.

Communication network for Smart Grid measurement and monitoring devices.

Smart Grid Communications and Standards.

IoT applications in Smart Grid.

Technology Transfer

Synergy Systems & Solutions Pvt. Ltd	A SOC based unique phasor processing architecture for micro-Phasor Measurement (u-PMU) device.	Transferred✓
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Patents

Indian Patent	An Apparatus for Micro Resolution Phasor Measurement. (Patent. No. 410521)	Granted ✓
Indian Patent	Breaker Guard System and Method for Controlling Operations of Electrical Circuit Breakers Based on Authentication. (app. No. 202411068969)	Under Review
Indian Patent	A SCADA integrated data aggregation system and method thereof: (Patent. No. 587957)	Granted ✓
Indian Patent	Method and Apparatus for providing secure real-time sampled value communication in digital substation. (Patent. No. 603812)	Granted ✓
Indian Patent	Machine Learning (ML)-Based System and Method for Microgrid Fault Detection Using Current Measurements (app. No. 202511120464)	Under Review

Transactions and Journals

IEEE IAS (Author)	A Data Acquisition Framework for Synchrophasor Measurement Applications in Distribution Grid	Under Review
IEEE TIM (Author)	Enhanced Power Distribution Monitoring using Compressed PoW Measurements Based on Sparse Decomposition.	Accepted ✓
IEEE TII (Author)	A Novel Processing Architecture for High Resolution Data Acquisition and Interpolated Dynamic Phasor Estimation	Under Review
IEEE IAS (Co-Author)	Edge-Driven Distributed Control for Power Distribution: A Real-Time Hardware-in-the-Loop Testbed for Industrial Automation and Applications DOI: 10.1109/TIA.2026.3678576	Accepted ✓
IEEE TIM (Author)	Frequency Adaptive Synchrophasor Estimation Using Multi DFT Bin Signal Model for Phasor Measurement Units	Under Review

Conferences

PESGM 2025 (Author) (Austin, Texas, USA)	Low Frequency Monitoring in Railways Electrical Network DOI: 10.1109/PESGM52009.2025.11225634	Accepted ✓
SEFET 2025 (Author) (Jaipur, Rajasthan, India)	A Unified Protocol Converter Framework for Integrating PMUs with SCADA Systems DOI: 10.1109/SEFET65155.2025.11255078	Accepted ✓
STPEC 2025 (Author) (Goa, India)	IoT Node Synchronized with Indian Constellation for Phasor Measurement in Power System DOI: 10.1109/STPEC66316.2025.11490975	Accepted  best paper
SGSMA 2026 (Author) (Santiago, Chile)	Evaluation of 5G communication for real time monitoring of power system using PMUs. DOI: 10.1109/SGSMA65955.2026.11618861	Accepted ✓
NPSC 2026 (Co-Author) (Patne, Bihar, India)	A Protocol Converter Framework for Integrating SunSpec Compliance Devices with SCADA.	Accepted ✓

Projects

1. Assessment of Abnormal Voltage Fluctuation Between Panki-Chakeri Railway Section of Prayagraj Division. (2024)

Work Order No: 230-Elect/TRD/PRYJ/32-A/53/731.

An experience of development and deployment of hardware PoW framework/prototype in Field:

A synchronized Point-on-Wave (PoW) Device is developed and employed for monitoring of railways electrical network. It continuously monitored the network for six months. The collected POW data later analysed for characterising the detected anomaly.

Skills:

Hands on experience of Embedded System FPGAs, Raspberry-pi, SOCs and other embedded hardware.

Expertise in programming languages: C, C++, Python, Assembly, VHDL and MATLAB

Hands on experience with HIL using: RTDS and Typhoon HIL,

Command on Designing and tinkering tools: Fusion 360 (AutoCAD), Altium (PCB designing)

Experience of Linux OS and Networking tools (N-map, Wireshark, Mininet etc).

Understanding of smart grid communication protocols (IEEE C37.118, DLMS, DNP3, IEC 61850, IEC 60870)

